

10/1/2024. Some derivation for CG :

Gr-S Orthogonalization process.

To understand why this process produced conjugate directions, let's consider the first iteration. Suppose d^0 is fixed:

$$d^1 = -g^1 + \frac{(g^1)^T d^0}{(d^0)^T d^0} d^0$$

We will show: $(d^1)^T d^0 = 0$

$$\begin{aligned} & (d^1)^T d^0 \\ &= -(g^1)^T d^0 + \frac{(g^1)^T d^0}{(d^0)^T d^0} (d^0)^T d^0 \\ &= -(g^1)^T d^0 + (g^1)^T d^0 = 0 \end{aligned}$$

The rest of iterations can be shown similarly

Exact Line Search.

To show how to find the optimal step size based on exact line search, see belows:

we need to show: $\alpha^* = \frac{d^T(Qx - b)}{d^T Q d}$

$$\alpha^* = \arg\min f(x + \alpha d)$$

optimality condition: $d^T \nabla f(x + \alpha^* d) = 0$ (*)

Note that: $f = \frac{1}{2} x^T Q x - b^T x$

$\Rightarrow \nabla f = Qx - b$. plug into (*)

$$d^T [Q(x + \alpha^* d) - b] = 0$$

$$\Rightarrow \alpha^* = \frac{d^T (b - Qx)}{d^T Q d}$$